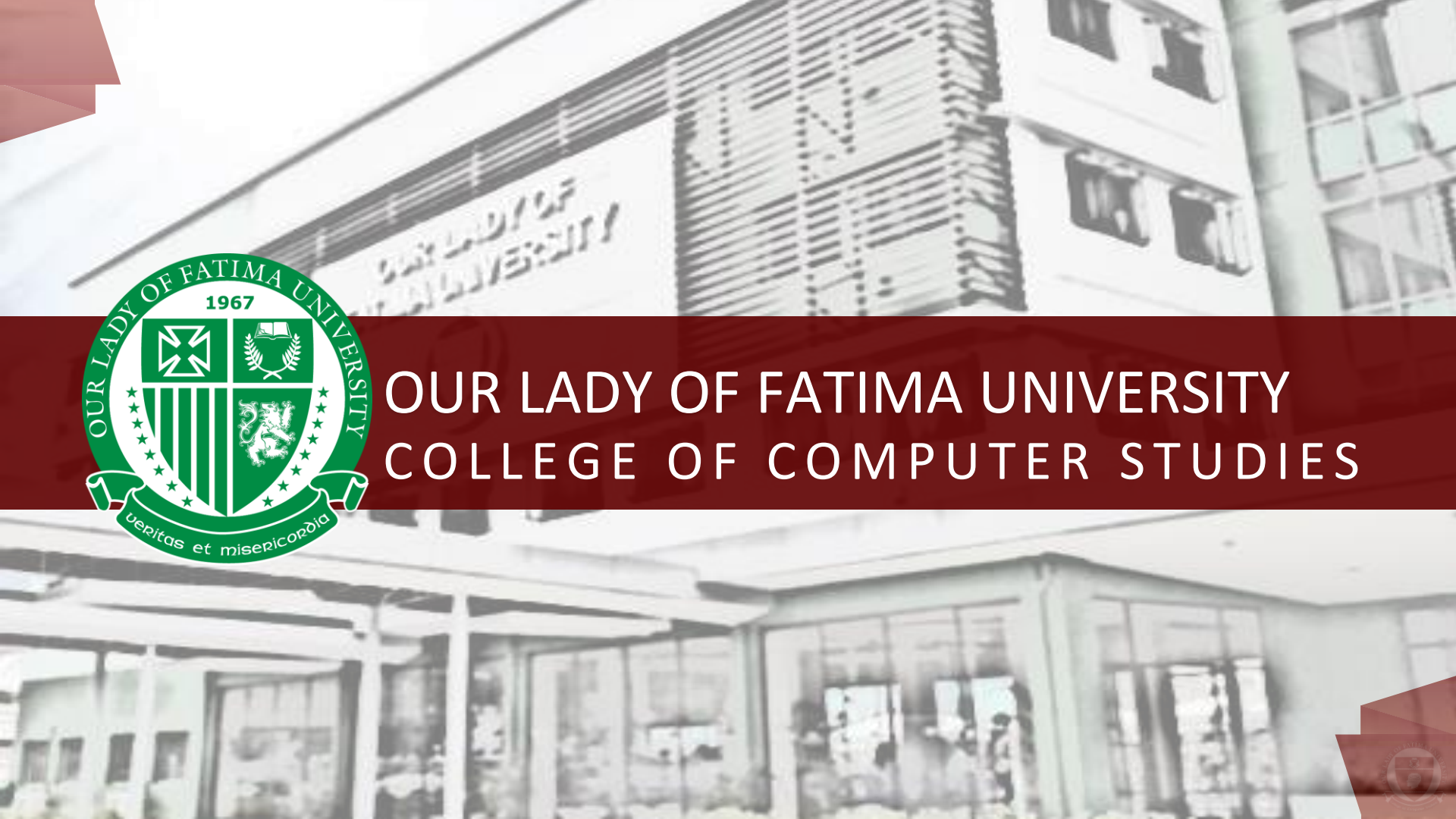
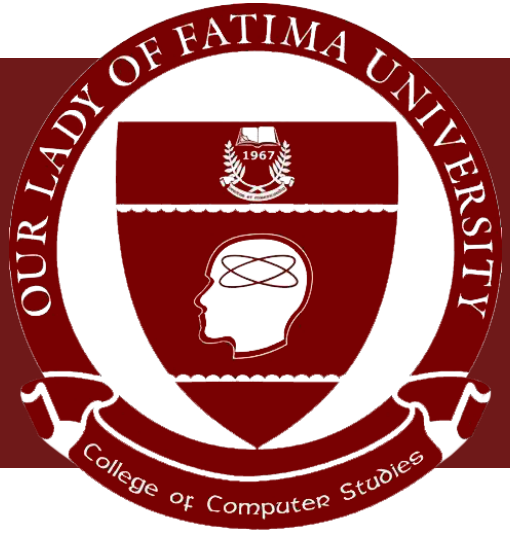
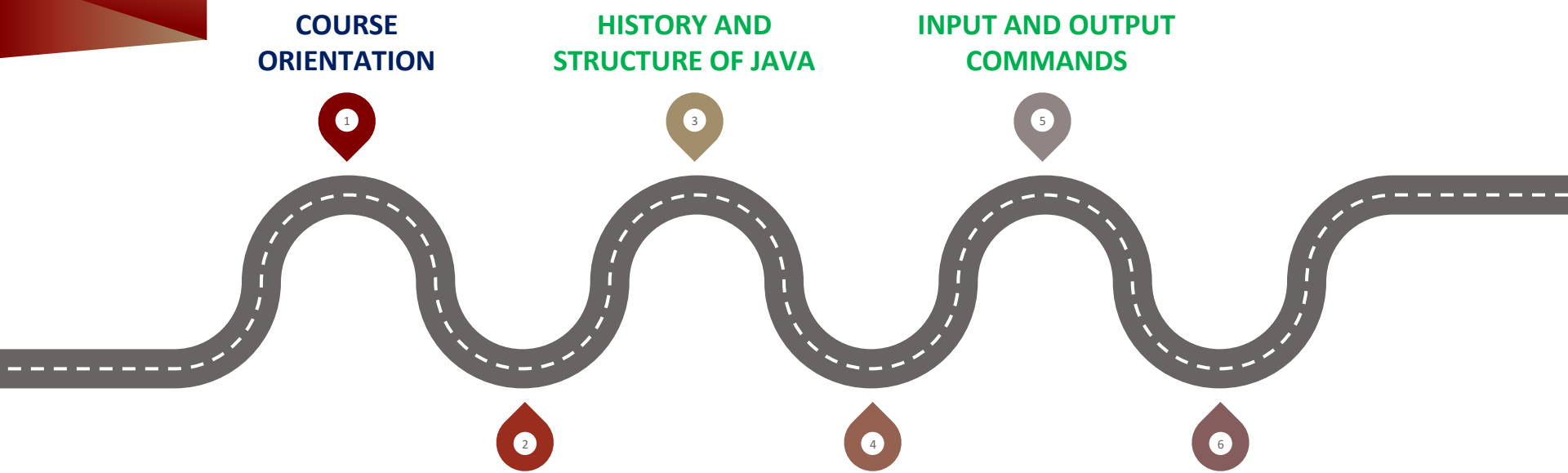


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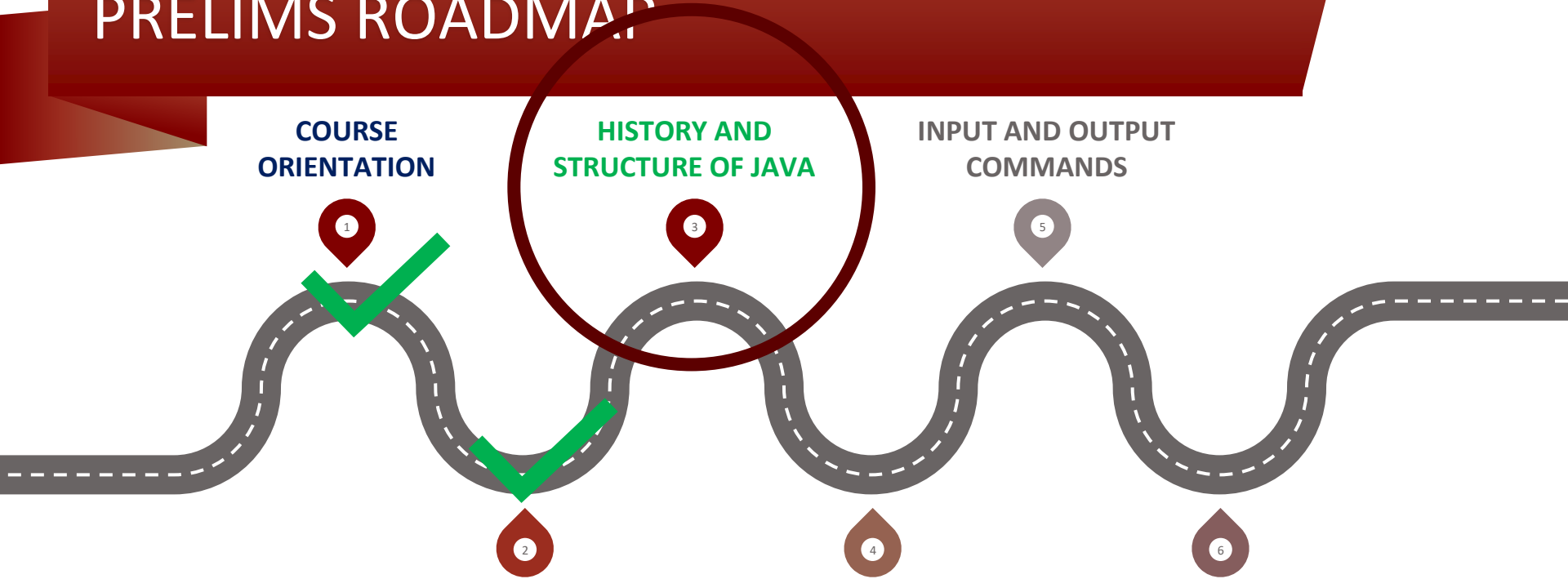


OOPR211
OBJECT-ORIENTED
PROGRAMMING I

PRELIMS ROADMAP



PRELIMS ROADMAP



Module 2: HISTORY AND STRUCTURE OF JAVA

Module Objectives:

1

Familiarize the structure of Java Language

2

Learn to compile a basic program using Java Syntax

Topics:

- History of Java
- Features of Java
- Structure of a Java Program
- Programming Errors in Java

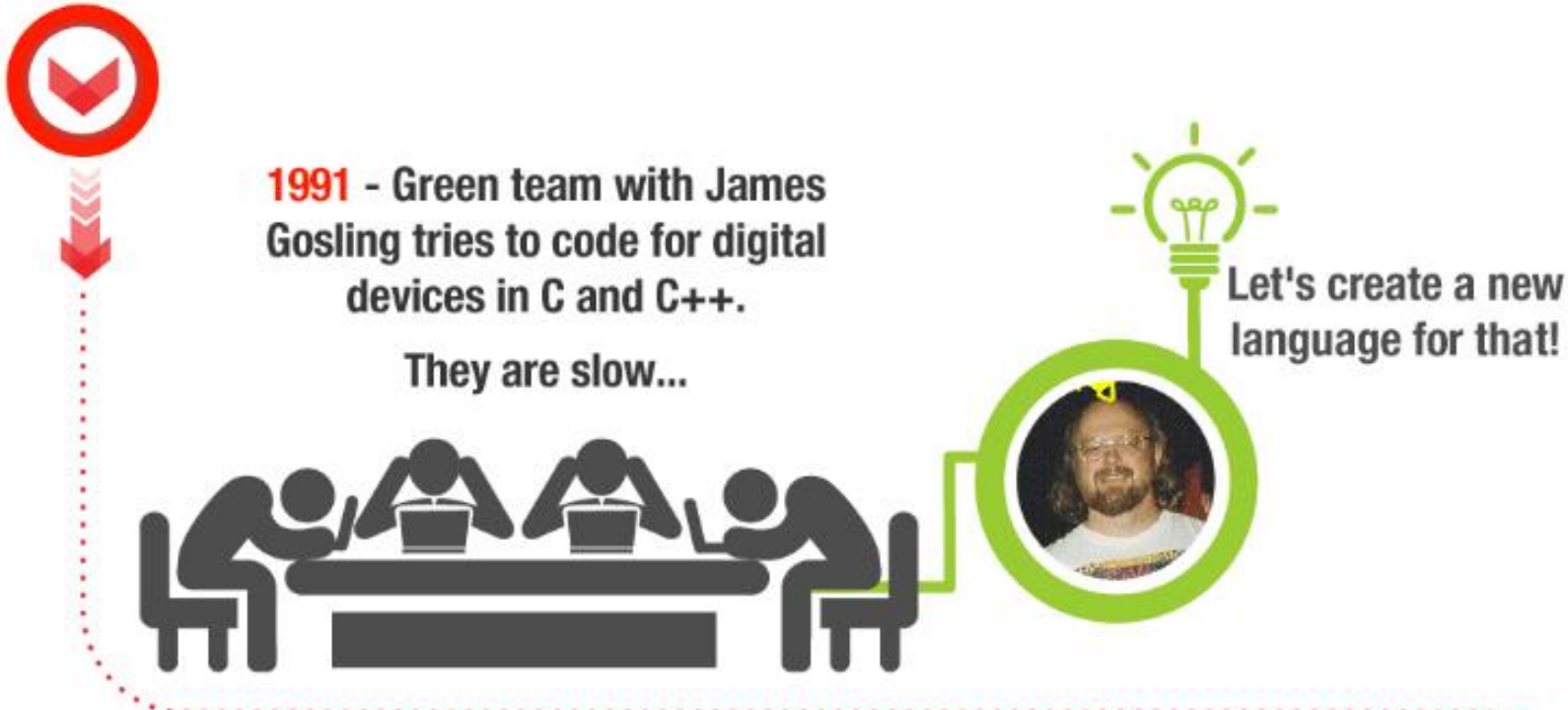
1 History of Java

It begins in 1991...



the company that introduce Java

It begins in 1991...



THE Green Team





GREEN PROJECT

A research project that leads to the development of Java.

It was intended for consumer electronic devices such as televisions and computers.



JAMES GOSLING

CHIEF ENGINEER of the Green Project



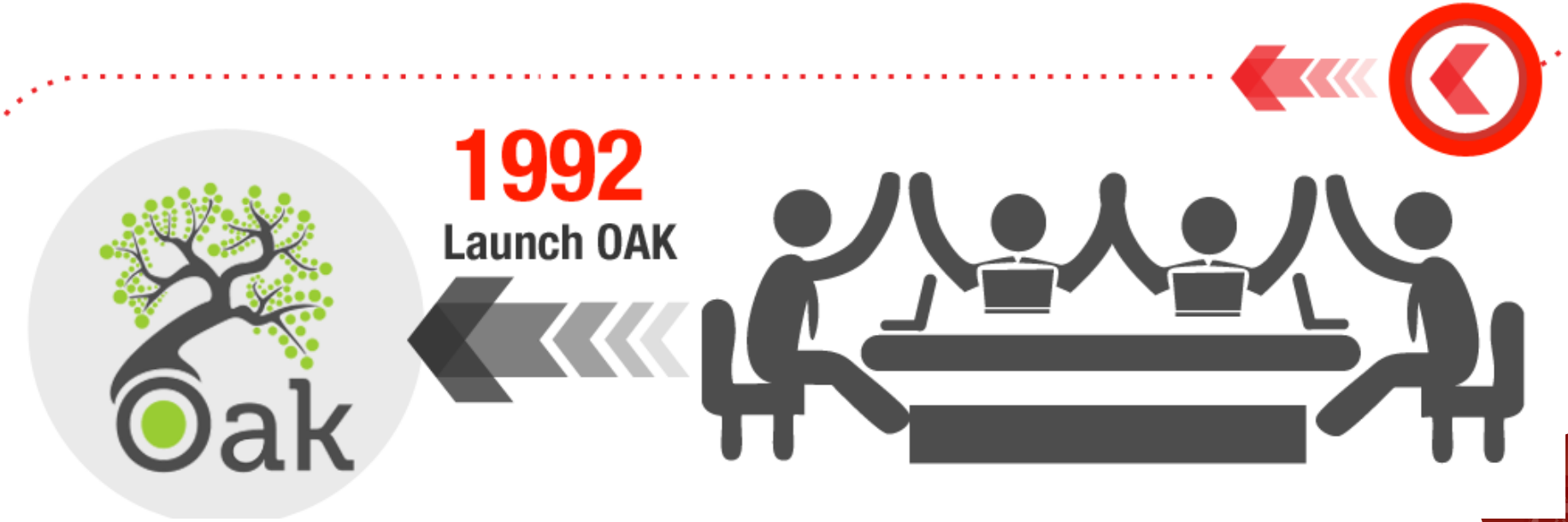
JAMES GOSLING

created the

OAK PROGRAMMING LANGUAGE

to meet the objective of green project.

In 1992...



OAK

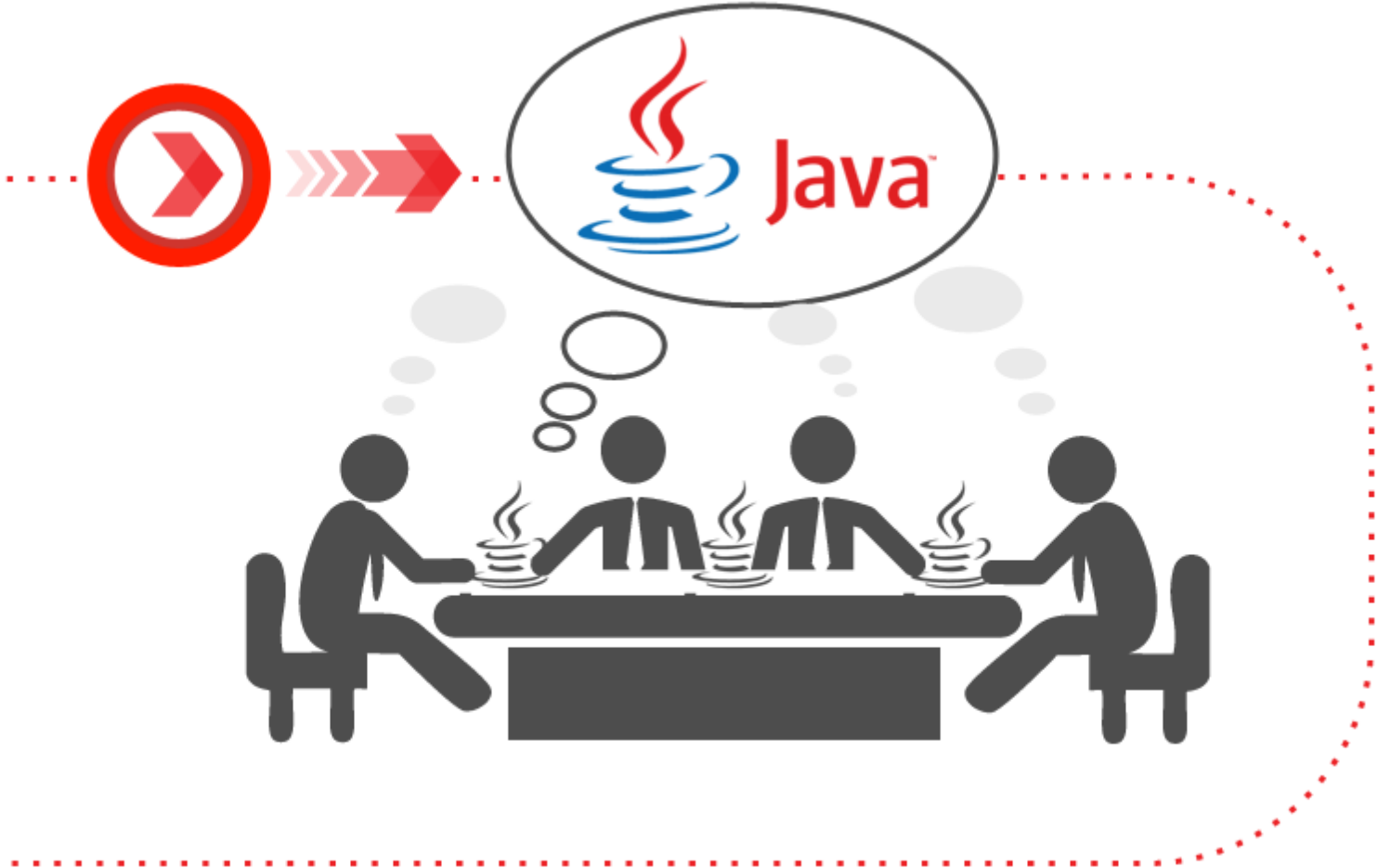
FIRST NAME OF JAVA WAS NAMED AFTER AN OAK TREE OUTSIDE HIS OFFICE.



In 1992...



oak.. no.. no..


Java

JAVA

DERIVED FROM LOCAL COFFEE SHOP.

THAT'S WHY ITS ICON IS LIKE A "CUP OF COFFEE".



In 1995...



Java™ Is officially Born.

In 1995...



Java
is officially born


Java
is free
open source

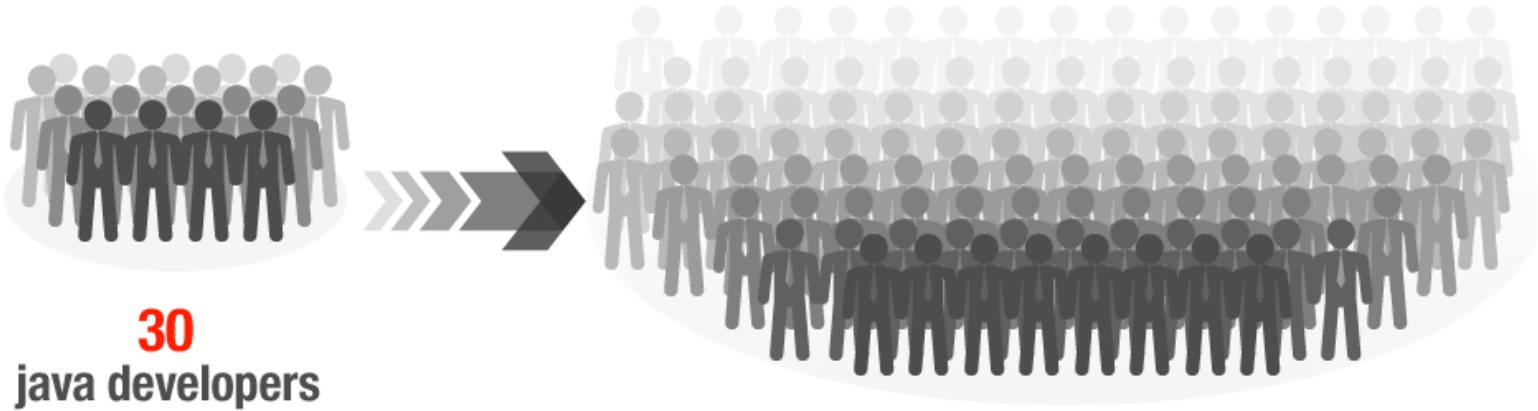
The PROMISE...

WORA

"Write Once, Run Anywhere"



Today...



30
java developers
in 1995

9 million
today

In 2010...





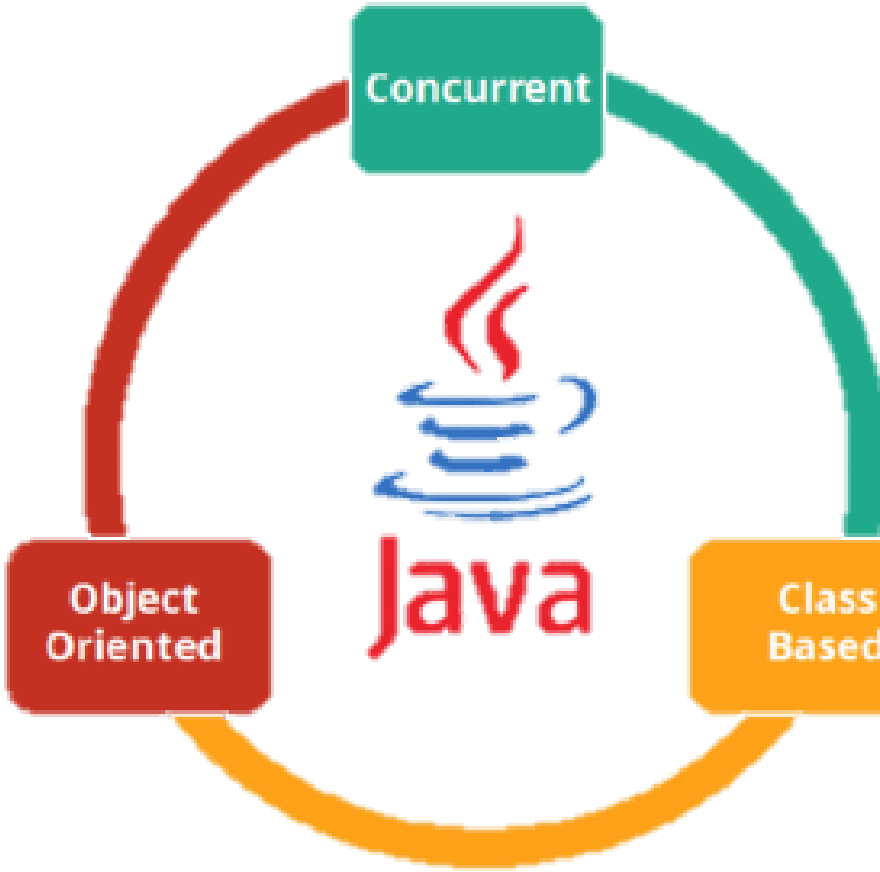


WHAT IS JAVA



JAVA

A GENERAL-PURPOSE, CONCURRENT, OBJECT-ORIENTED, CLASS-BASED, AND THE RUNTIME ENVIRONMENT(JRE) WHICH CONSISTS OF JVM WHICH IS THE CORNERSTONE OF THE JAVA PLATFORM



JAVA

AN OBJECT-ORIENTED LANGUAGE FOR GENERAL-PURPOSE BUSINESS APPLICATIONS AND FOR INTERACTIVE, WEB-BASED INTERNET APPLICATIONS.

JAVA IS FREE TO ACCESS AND CAN RUN ON ALL PLATFORMS.

WHAT IS JAVA USED FOR



What is Java Used for?

BANKING:

TO DEAL WITH TRANSACTION MANAGEMENT.

RETAIL:

BILLING APPLICATIONS THAT YOU SEE IN A STORE/RESTAURANT ARE COMPLETELY WRITTEN IN JAVA.

INFORMATION TECHNOLOGY:

JAVA IS DESIGNED TO SOLVE IMPLEMENTATION DEPENDENCIES.

What is Java Used for?

ANDROID:

APPLICATIONS ARE EITHER WRITTEN IN JAVA OR USE JAVA API.

FINANCIAL SERVICES:

IT IS USED IN SERVER-SIDE APPLICATIONS.

STOCK MARKET:

TO WRITE ALGORITHMS AS TO WHICH COMPANY THEY SHOULD INVEST IN.

What is Java Used for?

BIG DATA:

HADOOP MAPREDUCE FRAMEWORK IS WRITTEN USING JAVA.

SCIENTIFIC AND RESEARCH COMMUNITY:

TO DEAL WITH HUGE AMOUNT OF DATA.

2

Features of Java

FEATURES OF JAVA



SIMPLE

JAVA HAS MADE LIFE EASIER BY REMOVING ALL THE COMPLEXITIES SUCH AS POINTERS, OPERATOR OVERLOADING AS YOU SEE IN C++ OR ANY OTHER PROGRAMMING LANGUAGE.

FEATURES OF JAVA



PORTABLE

JAVA IS PLATFORM INDEPENDENT WHICH MEANS THAT ANY APPLICATION WRITTEN ON ONE PLATFORM CAN BE EASILY PORTED TO ANOTHER PLATFORM.

FEATURES OF JAVA



OBJECT-ORIENTED

EVERYTHING IS CONSIDERED TO BE AN “OBJECT” WHICH POSSESS SOME STATE, BEHAVIOR AND ALL THE OPERATIONS ARE PERFORMED USING THESE OBJECTS.

FEATURES OF JAVA





SECURED

ALL THE CODE IS CONVERTED IN BYTECODE AFTER COMPILATION, WHICH IS NOT READABLE BY A HUMAN. AND JAVA DOES NOT USE AN EXPLICIT POINTER AND RUN THE PROGRAMS INSIDE THE SANDBOX TO PREVENT ANY ACTIVITIES FROM UNTRUSTED SOURCES.

IT ENABLES TO DEVELOP VIRUS-FREE, TAMPER-FREE SYSTEMS/APPLICATIONS.

FEATURES OF JAVA





DYNAMIC

DYNAMIC: IT HAS THE ABILITY TO ADAPT TO AN EVOLVING ENVIRONMENT WHICH SUPPORTS DYNAMIC MEMORY ALLOCATION DUE TO WHICH MEMORY WASTAGE IS REDUCED AND PERFORMANCE OF THE APPLICATION IS INCREASED.

FEATURES OF JAVA



DISTRIBUTED

JAVA PROVIDES A FEATURE WHICH HELPS TO CREATE DISTRIBUTED APPLICATIONS. USING REMOTE METHOD INVOCATION (RMI), A PROGRAM CAN INVOKE A METHOD OF ANOTHER PROGRAM ACROSS A NETWORK AND GET THE OUTPUT.

FEATURES OF JAVA

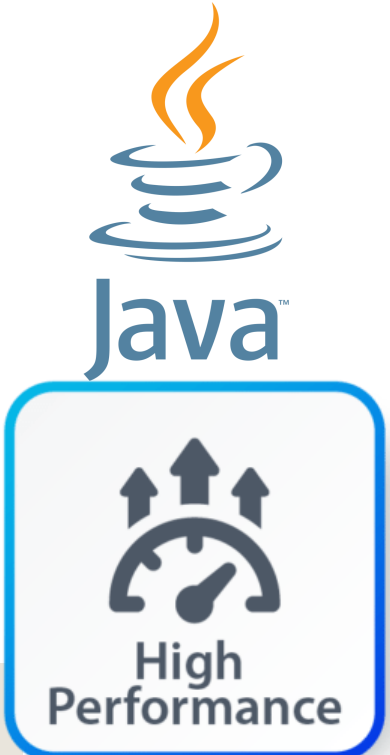




ROBUST

ROBUST: JAVA HAS A STRONG MEMORY MANAGEMENT SYSTEM. IT HELPS IN ELIMINATING ERROR AS IT CHECKS THE CODE DURING COMPILE AND RUNTIME.

FEATURES OF JAVA



HIGH PERFORMANCE

JAVA ACHIEVES HIGH PERFORMANCE THROUGH THE USE OF BYTECODE WHICH CAN BE EASILY TRANSLATED INTO NATIVE MACHINE CODE. WITH THE USE OF JIT (JUST-IN-TIME) COMPILERS, JAVA ENABLES HIGH PERFORMANCE.

FEATURES OF JAVA



INTERPRETED

INTERPRETED: JAVA IS COMPILED TO BYTECODES, WHICH ARE INTERPRETED BY A JAVA RUN-TIME ENVIRONMENT.

FEATURES OF JAVA



MULTITHREADED

JAVA SUPPORTS MULTIPLE THREADS OF EXECUTION (A.K.A., LIGHTWEIGHT PROCESSES), INCLUDING A SET OF SYNCHRONIZATION PRIMITIVES. THIS MAKES PROGRAMMING WITH THREADS MUCH EASIER.

3

Structure of a Java Program

Basic Structure of JA

```
public class First
{
    public static void main(String[] args)
    {
        System.out.println("First Java application");
    }
}
```

CLASS DEFINITION

```
public class [ClassName]
        {  // Begin Class
                        // Class Body
        } // End Class
```

IT CONTAINS ALL THE PIECES OF THE PROGRAM NECESSARY TO EXECUTE THE PROGRAM.

CLASS DEFINITION

```
public class First
{
    public static void main(String[] args)
    {
        System.out.println("First Java application");
    }
}
```

Class name

class body

Class Body – is where attributes are declared and methods are defined.

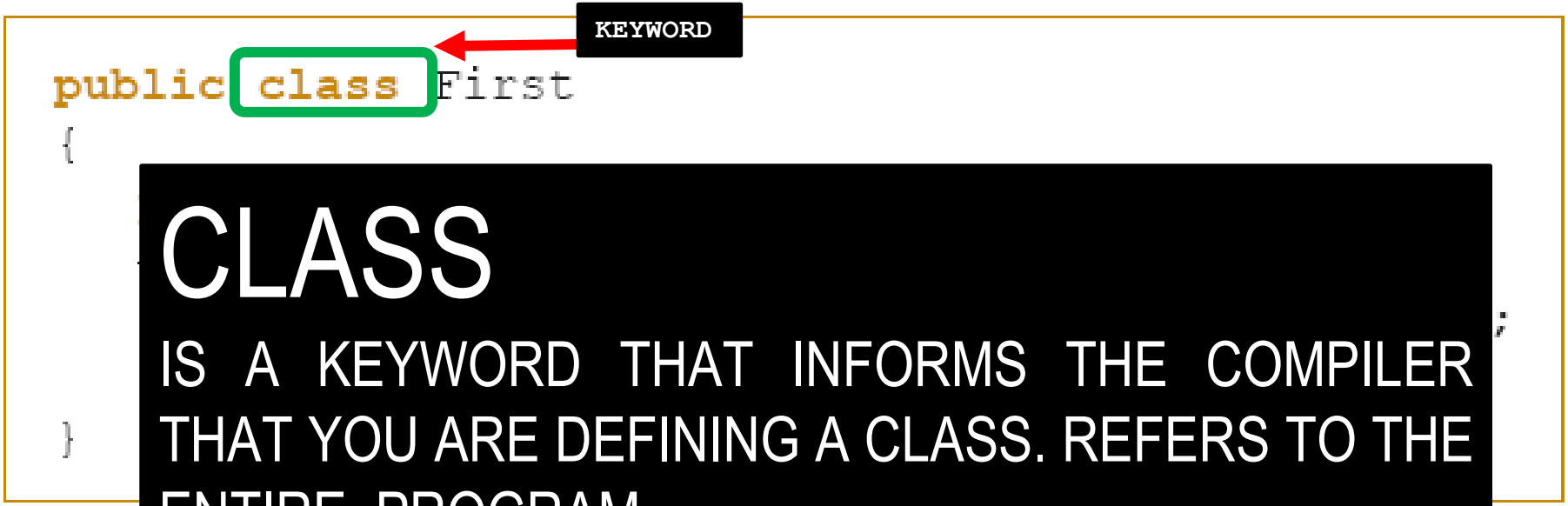
CLASS DEFINITION



PUBLIC – IS AN ACCESS MODIFIER THAT INDICATES THAT THE CODE COULD BE ACCESSED BY ALL OBJECTS.

ACCESS MODIFIER - SPECIFIES THE CIRCUMSTANCES IN WHICH CLASS COULD BE ACCESSED.

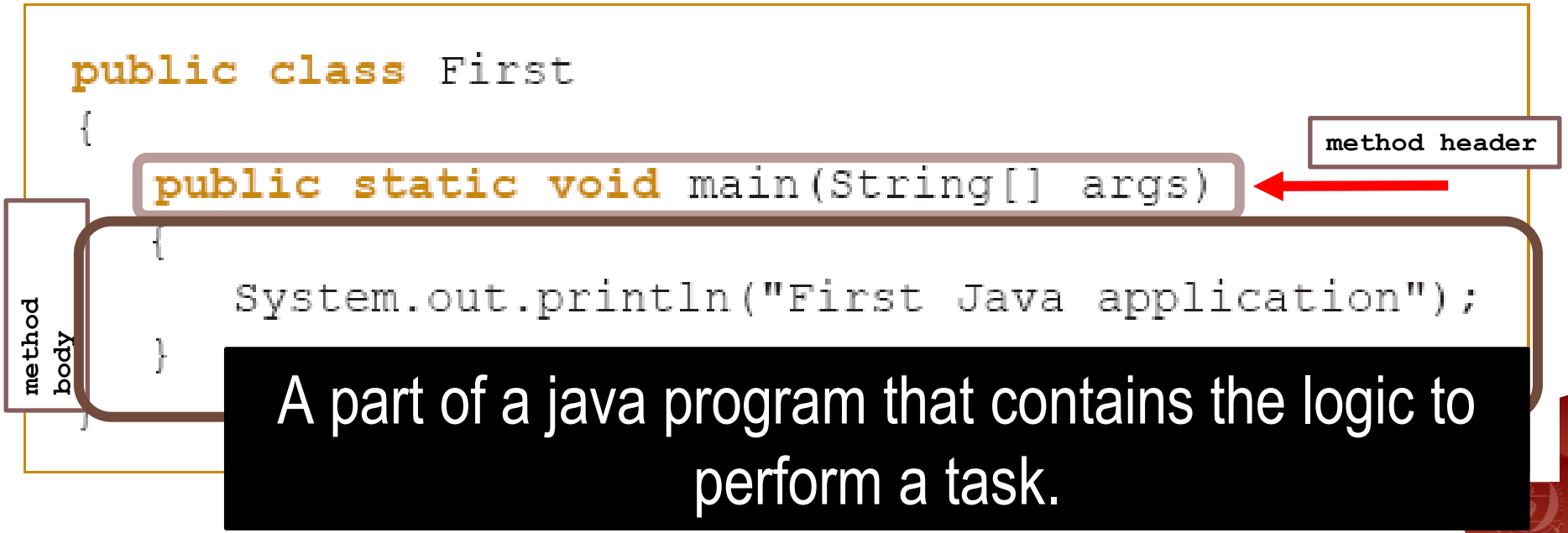
CLASS DEFINITION



CLASS

IS A KEYWORD THAT INFORMS THE COMPILER THAT YOU ARE DEFINING A CLASS. REFERS TO THE ENTIRE PROGRAM.

METHOD DEFINITION



A part of a java program that contains the logic to perform a task.

METHOD DEFINITION

```
public class First
{
    public static void main(String[] args)
    {

    }
}
```



Method Modifier – enables setting of properties for the method. Means that the method is for class.

METHOD DEFINITION

```
public class First
{
    public static void main(String[] args)
    {
        System.out.println("
    }
}
```

void – means that the method does not return a value when it is called.

METHOD DEFINITION

```
public class First
{
    public static void main(String[] args)
    {
        System.out.println("First Java app
    }
}
```

Method Name

Method Argument List

String – refers to a data type.
args – identifier to any string character.

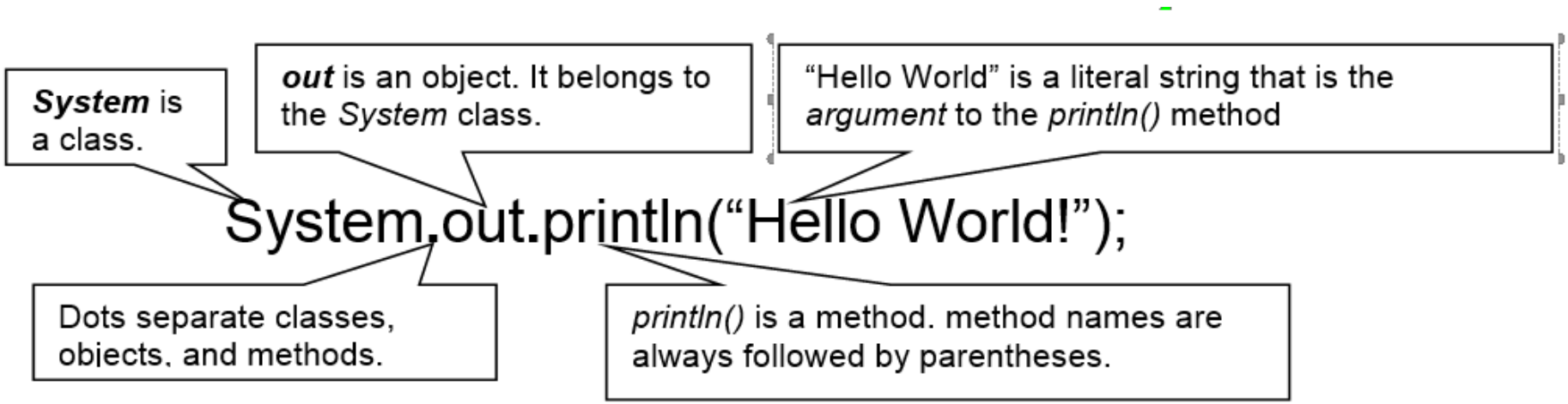
“Hello World Program”

Example 1: (Printing a simple text)

```
public class Hello
        {
                public static void main( String args [ ] )
                        {
                        System.out.println(“Hello World!”);
                        }
        }
```

```
Hello World!
```

“Hello World Program”



Welcome to Java Programming

Example 2: : (Multiple line of text)

```
public class Welcome
        {
                public static void main( String args [] )
                        {
                        System.out.println ("Welcome\nto\nJava\nProgramming!");
                        }
        }
```

```
Welcome
to
Java
Programming!
```

\

- CALLED AS ESCAPE CHARACTER
- WHICH INDICATES THAT A SPECIAL CHARACTER IS TO BE OUTPUT.

ESCAPE SEQUENCE

\n	• newline
\t	• horizontal tab
\\	• backslash
\”	• double quote

COMMENTS

// - Single line comment

/ */ - Multiple-line comment*

COMMENTS

```
/*
Laboratory Exercise 1: Printing my name and address on the screen
Programmer: Lailanie M. Calma
Date:  September 13, 2021
Program Name: Address
*/

public class Address
{
public static void main (String[] args)//method header
{
```

4

Programming Errors in Java

PROGRAMMING ERRORS

IN COMPUTER TERMS AN ERROR IS A BUG THAT AFFECTS THE OUTPUT OF A PROGRAM. IT MAY ALSO SIMPLY MEAN AN INCORRECT STATEMENT WITHIN A PROGRAM.

SYNTAX ERRORS

THESE ARE USUALLY TYPING ERRORS

SYNTAX – REFERS TO CODE STRUCTURE OR GRAMMAR.

SEMANTIC ERRORS

UNRECOGNIZABLE CODE TO THE JAVA COMPILER.

RUN-TIME ERRORS

ERRORS DISPLAYED DURING PROGRAM EXECUTION

Error	Error Message	Method of Correction
Missing semi colon	Invalid type expression Invalid declaration ';' expected	Add a semi colon to the end of the line.
Missing punctuation	') ' expected	Insert missing) or }
Incorrect number of arguments	Invalid argument	Add a comma between arguments in an argument list
Incorrect file name	Public class must be define in a file	Make sure that class name and file name are the same in spelling and case
Incorrect usage of mathematical operator	Missing term	Correct operand error

QUESTIONS?

Thank you!

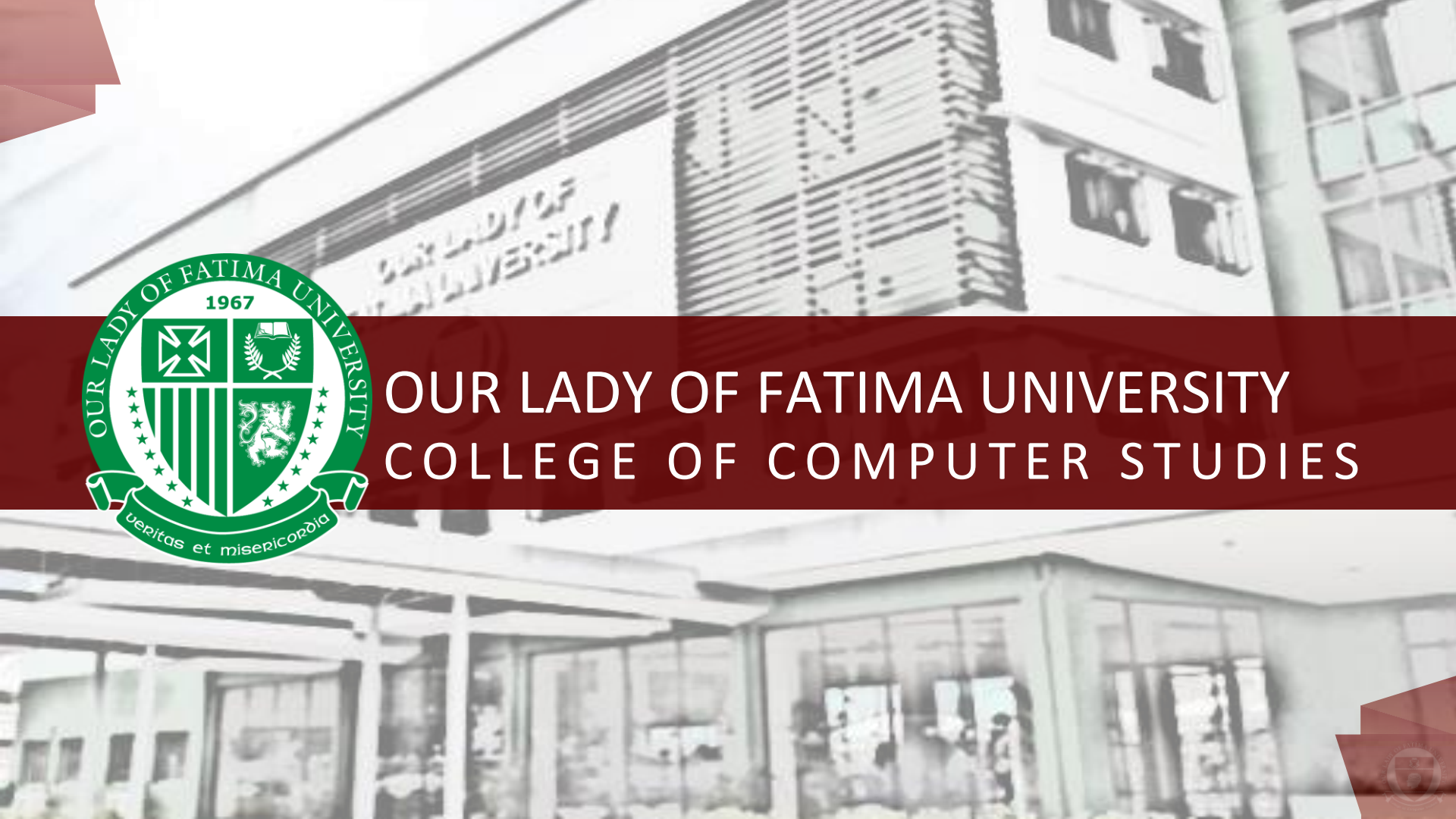
Credits

Special thanks to all the people who made and released these awesome resources for free:

- Presentation template by SlidesCarnival
- Photographs by Unsplash

Additional References:

- https://www.edureka.co/blog/what-is-java/
- http://ecomputernotes.com/java/what-is-java-language/types-of-java-programs



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